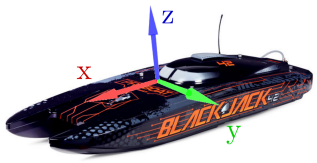
Solutions: Transfer Functions and Modeling



USV Model

Parameters: $m = 5.0\,\text{kg}$, $b_x = 8.0\,\text{N}\cdot\text{s/m}$, $I = 0.8\,\text{kg}\cdot\text{m}^2$, $b_r = 1.2\,\text{N}\cdot\text{m}\cdot\text{s/rad}$, $N_\delta = 2.0\,\text{N}\cdot\text{m/rad}$.

6. Surge equation of motion (Newton's second law along x, thrust minus linear drag):

$$\boxed{m\,\dot{v}_x(t) + b_x\,v_x(t) = F(t)} \qquad \Rightarrow \qquad 5.0\,\dot{v}_x + 8.0\,v_x = F$$

   Dividing by b_x puts it in standard first-order form, which identifies the time constant and DC gain:

$$\underbrace{\frac{m}{b_x}}_{\tau}\,\dot{v}_x + v_x = \underbrace{\frac{1}{b_x}}_{K_{dc}}\,F \qquad \Rightarrow \qquad 0.625\,\dot{v}_x + v_x = 0.125\,F$$

   Color key: input amplitude, DC gain, time constant.

7. With $v_x(0) = 0$ and a step of amplitude $F_0 = 12\,\text{N}$, $F(s) = F_0/s$:

$$V_x(s) = \frac{K_{dc}F_0}{s(\tau s + 1)} = \frac{0.125 \cdot 12}{s(0.625\,s + 1)} = \frac{1.5}{s} - \frac{1.5}{s + 1/0.625}$$

$$\boxed{\begin{aligned} v_x(t) &= K_{dc}F_0\left(1 - e^{-t/\tau}\right) = (0.125 \cdot 12)\left(1 - e^{-t/0.625}\right)\ \text{m/s} \\ &= 1.5\left(1 - e^{-t/0.625}\right)\ \text{m/s} \end{aligned}}$$

   - Plot: first-order rise from 0 toward 1.5 m/s with time constant $\tau = m/b_x = 0.625\,\text{s}$. (Both mass and drag set the time constant, i.e., how quickly the vessel accelerates.)
   - Steady-state speed: $v_{ss} = K_{dc}F_0 = 0.125 \cdot 12 = 1.5\,\text{m/s}$. (Top speed depends only on drag, $K_{dc} = 1/b_x$, not on mass.)
   - 95% of steady state at $t \approx 3\tau \approx 1.9\,\text{s}$.

8. Surge transfer function, thrust to surge velocity:

$$\boxed{\begin{aligned} G_x(s) = \frac{V_x(s)}{F(s)} &= \frac{1}{ms + b_x} = \frac{1/b_x}{\frac{m}{b_x}s + 1} = \frac{K_{dc}}{\tau s + 1} \\ &= \frac{0.125}{0.625\,s + 1} = \frac{1}{5.0\,s + 8.0} = \frac{0.2}{s + 1.6} \end{aligned}}$$

   - Units: V_x/F in (m/s)/N.
   - Pole at $s = -1/\tau = -1.6\,\text{s}^{-1}$.
   - DC gain $K_{dc} = 1/b_x = 0.125$ (m/s)/N.

9. Yaw equation of motion (moment balance about z, rudder moment minus linear drag):

$$\boxed{I\,\dot{r}(t) + b_r\,r(t) = N_\delta\,\delta(t)} \qquad \Rightarrow \qquad 0.8\,\dot{r} + 1.2\,r = 2.0\,\delta$$

10. Yaw-rate transfer function, rudder to yaw rate:

$$\boxed{\begin{aligned} G_r(s) = \frac{R(s)}{\Delta(s)} = \frac{N_\delta}{Is + b_r} &= \frac{\overbrace{N_\delta/b_r}^{K_{dc}}}{\underbrace{(I/b_r)}_{\tau}\,s + 1} = \frac{K_{dc}}{\tau s + 1} \\ &= \frac{1.67}{0.667\,s + 1} = \frac{2.0}{0.8\,s + 1.2} = \frac{2.5}{s + 1.5} \end{aligned}}$$

- Units: R/Δ in (rad/s)/rad.
- Time constant $\tau = I/b_r = 0.8/1.2 = 0.667\,\mathrm{s}$; pole at $s = -1/\tau = -1.5\,\mathrm{s}^{-1}$.
- DC gain $K_{dc} = N_\delta/b_r = 2.0/1.2 = 1.67$ (rad/s)/rad.

11. Heading transfer function. Since $r = \dot{\psi}$, $R(s) = s\,\Psi(s)$ (zero initial heading), so heading adds an integrator:

$$\boxed{G_\psi(s) = \frac{\Psi(s)}{\Delta(s)} = \frac{N_\delta}{s(Is + b_r)} = \frac{2.0}{s(0.8\,s + 1.2)} = \frac{2.5}{s(s + 1.5)}}$$

Units: Ψ/Δ in rad/rad; poles at $s = 0$ and $s = -1.5\,\mathrm{s}^{-1}$.

12. With $\Delta(s) = 0.3/s$, the final value theorem gives

$$r_{ss} = \lim_{s\to 0} s\,G_r(s)\,\frac{0.3}{s} = \frac{2.5}{1.5}(0.3) \quad \Rightarrow \quad \boxed{r_{ss} = 0.5\,\mathrm{rad/s}}$$

(about 28.6 deg/s). The heading $\psi(t)$ grows without bound, so the vessel turns in a circle.